

Markscheme

Mathematics: analysis and approaches
Practice question bank

165. (a)

The five roots of $z^5 = 1$ are the 5th roots of unity.

Find the sum of all five roots, and hence find the sum of the four non-real roots.

the sum of the n th roots of unity is 0 for $n \geq 2$ (coefficient of z^{n-1} in $z^n - 1$ is zero) *(M1)* **R1**

one root is the real root $z = 1$ *(M1)*

sum of non-real roots = $0 - 1$ **A1**

$$= -1$$

[5 marks]